

Reading the H–R Diagram *by yourself*

An astronomer hands you a chart with thousands of dots and four axes. By the end of this lesson, you should be able to read it like a map. Use the simulation to work things out — answer in your own words, in the spaces below.

Time: 45 min Tool: Interactive H–R Diagram simulation

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1 Working out *what the axes mean*

~ 8 MIN

Open the simulation in *Explore* mode. The gold star is yours to drag. Don't worry about the regions or the real stars yet — those toggles can come off if they're distracting.

1

NOTICE

Drag your star slowly to the left along the bottom of the diagram. What happens to the temperature reading in the side panel? Now drag it slowly to the right. Write down the rule.

→ Action: drag horizontally and watch the Temperature value

2

NOTICE

Now drag your star slowly upwards on the diagram. Look at the Luminosity reading. By how much does it change as you cross one major gridline? Try crossing two or three.

→ Action: drag vertically and read the Luminosity values at major gridlines

3

COMPARE

Look at the right-hand axis: Absolute Magnitude (M). Drag your star to where Luminosity = $1 L_{\odot}$. What does M read? Now drag to where $L = 100 L_{\odot}$. Now to $L = 0.01 L_{\odot}$.

Record the M value at each of those luminosities, then describe the relationship between L and M. In particular: *is a brighter star a bigger or smaller M?*

4

INSPECT

Find the **Spectral Class** reading in the side panel. As you drag your star left to right, the letter changes. **Write down the order of letters from hottest (left) to coolest (right).**

PULL IT TOGETHER

*Without looking at the simulation, write down the four conventions of an H–R diagram. **Pretend you're explaining them to someone who's never seen one.** What's "weird" about each axis compared to a normal graph?*

✓ Tell me when you're at this point — I'll come over and check your answers before you move on.

2 Mapping *where stars actually live*

~ 10 MIN

Stars aren't scattered randomly across the diagram — they cluster in specific zones. Turn on *Show real stars* to see this. Each named star is a real one in our sky.

5

OBSERVE

Just look at the diagram with the real stars showing. Don't drag anything yet. Where do most of the named stars sit? Are they evenly scattered, or do you see clusters?

6

INVESTIGATE

Use the **Jump to a star** buttons to visit each of these stars. For each, record T and L, and write what kind of star you'd guess it is just from its position on the diagram:

Sun

Sirius B

Sirius A

Aldebaran

Betelgeuse

7

CONFIRM

Now turn on **Show regions**. The diagram lights up with four shaded zones. **Did your guesses for question 6 match the regions?** Were any wrong — and if so, why might you have got them wrong?

8

DEFINE

For each of the four regions, write a **one-sentence definition** in your own words, focusing on temperature and luminosity:

Main sequence

Red giants

Supergiants

White dwarfs

PULL IT TOGETHER

Most stars sit on the main sequence. Why might that be? What does sitting on the main sequence tell you about what a star is currently doing?

3 Discovering the link between T , L , and R

~ 10 MIN

Each point on the diagram corresponds to a specific star *radius*, even though radius isn't an axis. Your job is to work out why. Turn on *Show isoradius lines* if it helps you see the pattern.

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EXPERIMENT

Drag your star straight up the diagram, keeping the temperature roughly constant (around 6000 K). Watch the radius reading. Watch the size of the star itself in the diagram. **What's happening?**

→ Action: drag from $L = 0.01$ up to $L = 10^4$ at the same T value

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EXPERIMENT

Now do the opposite: **drag your star horizontally**, keeping luminosity roughly constant. What happens to the radius this time?

→ Action: drag along a horizontal line at $L = 1000$ from hot to cool

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REASON

You should know the equation $L = 4\pi R^2 \sigma T^4$. Use what you've just observed (and a bit of algebra) to **predict** what would happen to the radius if you doubled both T and L . Then check your prediction in the simulation.

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APPLY

Rigel and Betelgeuse have very similar luminosities (both around $10^5 L_{\odot}$). Use the simulation to confirm this. Yet Rigel is hot and blue, while Betelgeuse is cool and red. **Use the equation $L = 4\pi R^2 \sigma T^4$ to explain how this is possible.** Be specific about which has the larger radius and roughly by how much.

PULL IT TOGETHER

If someone told you only a star's position on the H – R diagram, what would you immediately know about it? List as many things as you can.

✓ Wave me over here — I'd like to hear your answer to question 12 before you move on.

4 Following *the life of a Sun-like star*

~ 10 MIN

Switch to *Lifecycle: A-Level* mode at the top. Make sure mass is set to $1 M_{\odot}$. The star will move through the stages of its life as you click Next. Read each stage carefully — both the description and the "What happens next" box.

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TRACE

Step through all five stages. For each stage, record (a) where on the diagram the star sits, (b) what it's *doing* physically (what's fusing, what's contracting, etc.), and (c) what causes the star to leave that stage.

Stages: Stellar Nebula · Main Sequence · Red Giant · Planetary Nebula · White Dwarf

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IDENTIFY THE CENTRAL IDEA

One physical principle keeps appearing in the explanations of why each stage starts and ends. It's about a balance of two forces. Read the descriptions carefully and write down what this principle is, in your own words.

+ STUCK?

15

REASON

Using your answer to 14: explain **why a star can't just stay on the main sequence forever**. What makes the equilibrium fail?

PULL IT TOGETHER

Sketch the path the star takes across the H – R diagram during its lifecycle, labelling each stage. Then add a short note next to each transition explaining why it happens.

5 Mass *determines fate*

~ 10 MIN

Switch the mass selector through all three options: $1 M_{\odot}$, $15 M_{\odot}$, $30 M_{\odot}$. Run each lifecycle once through. Pay attention to how the path across the diagram differs.

16

COMPARE

For each mass ($1 M_{\odot}$, $15 M_{\odot}$, $30 M_{\odot}$), record:

- Where on the diagram does its main-sequence life begin?
- How long does it last? (*days, years, Myr, Gyr — order of magnitude is fine*)
- What's its biggest, brightest stage?
- How does it die?
- What's left behind at the end?

Don't just copy stage names — describe what's actually different about each star's life.

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SPOT THE RULE

Complete each statement in your own words:

- (a) The more massive a star is, the _____ its main sequence life.
- (b) Stars with mass less than about _____ $M_{\odot}$ end their lives as _____.
- (c) Stars with mass roughly between _____ and _____ $M_{\odot}$ end as _____.
- (d) Stars more massive than about _____ $M_{\odot}$ end as _____.

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PREDICT

Why might a more massive star *burn through its fuel faster*, even though it has more fuel? **Think about gravity, core temperature, and the rate of fusion reactions.**

+ STUCK?

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SYNTHESISE

A star is observed at $T \approx 25\,000\text{ K}$, $L \approx 50\,000 L_{\odot}$. Find this position on the diagram. **Predict** (and justify): (a) which region it's in, (b) what kind of end it will eventually have, and (c) roughly how its main-sequence life compares to the Sun's.

PULL IT TOGETHER

Looking back at everything you've worked out: what makes the H – R diagram such a powerful tool for an astronomer? Why is it more useful than just listing each star's properties in a table?

Before we regroup

If you've finished early: try predicting where Polaris would sit, then check using the Jump to a star button. Or try the Lifecycle: Curious mode for any of the masses — there are extra stages most undergraduate textbooks don't even cover.

If you got stuck somewhere, that's fine — we'll cover it as a class. Bring your questions to the plenary.